Maharashtra Engineering Training Academy, Nashik

Professional Exam Division

Practice Question for CEA

General Civil Engineering

सदर सराव प्रश्न हे परीक्षार्थींच्या सोईसाठी व सरावासाठी देण्यात येत आहेत. परीक्षार्थीच्या वारंवार मागणीनुसार सदर प्रश्नांची उत्तरे उपलब्ध करून देण्यात येतआहेत. तरी या सराव प्रश्न व उत्तरे यांची खात्री परीक्षार्थींनी आपल्या स्तरावर तपासून पाहणे. या विषयीच्या कुठल्याही तक्रारीची दखल या कार्यालयामार्फत घेण्यात येणार नाही याची नोंद घ्यावी.

Sr. no.	Question	Correct Option	Option 1	Option 2	Option 3	Option 4
1	---- is commonly used in sliding bearings	2	Crude Oil	Poly Tetra Fluoro Ethylene (PTFE)	Petrol	All of these
2	-------- means removal of soluble salts from the root zone	2	Clearing	Leaching	Cleaning	moulding
3	"Dead Slow" is a	1	regulatory sign	warning sign	informatory sign	none of these
4	1 hectare	4	2.01 acre	3.22 acre	2.71 acre	2.47 acre
5	1 mm rainfall over a hector = ------- cubic meter	3	200	100	10	25
6	1 ton of Portland cement equal to ---	2	10 bags	20 bags	30 bags	50 bags
7	A complete signal-time cycle constitutes	4	red timing	yellow timing	green timing	all of these
8	A facing of masonry or stones to protect an embankment from erosion.	2	abutment	revetment	deck	pier
9	A fluid which obeys the Newton's law of viscosity is termed as	3	Real fluid	Ideal fluid	Newtonian fluid	Non-Newtonian fluid
10	A member constructed of steel / concrete driven into the ground to carry axial loads	1	bearing pile	slab	pier	none of these
11	A piezometer opening in pipe measures	2	Velocity head	static pressure	total pressure	negative static pressure
12	A pilot tube is used to measure	3	pressure	differerence in pressure	velocity of flow	none of these

13	A pycnometer is used to determine	3	water content and voids ratio	specific gravity and dry density	water content and specific gravity	voids ratio and dry density
14	A rain gauge should preferably be fixed	3	near building	under tree	in an open space	in a closed space
15	A recording type rain gauge	4	produces a mass curve of rain fall	records the cumulative rain	is some times called integrating rain gauge or continuous rain gauge	all of these
16	A river training work is generally required when the river is	1	meandering	aggrading	degrading	all of these
17	A section with more reinforcement than balanced section is called	2	balanced	over reinforced	under reinforced	none of these
18	A solid core of rock is formed in side the cylinder in case of	3	auger boring	percussion drilling	diamond drilling	wash boring
19	A type of cast-in-situ pile which has an enlarged base and a corrugated stem, is	2	Simplex pile	Frankie pile	Vibro-pile	Raymond pile
20	A Venturimeter is used to measure	2	Pressure of liquid	Discharge of liquid	Pressure difference between two points in a channel	Pressure difference between two points in a pipe
21	A wooden block hinged on post outside a door is known	1	cleat	stop	horn	none of these
22	Addition of pozzolana to ordinary Portland cement increases	2	bleeding	shrinkage	permeability	heat of hydration
23	An arch may fail due to	4	Uneven settlement of abutments	Sliding of voussoirs	Crushing of the material	All of these

24	Area covering all floods collecting is called as ---	1	River Basin	Valley	Hills	All of these
25	ARG means	3	Automatic Random Gauge	Automatic Rain Guard	Automatic Rain Gauge	Average Rain Gauge
26	As per IRC specifications the camber for cement concrete surface roads is	1	1 in 60	1 in 80	1 in 100	1 in 120
27	Basalts in which practically no crystallization has taken place and which consist mostly of basalt glass are called	1	tachylytes	laterite	ignite	sand stone
28	Best type of piles for soft soil having little resistance to the flow of concrete, is	2	Simplex pile	Vibro pile	Raymond pile	Frankie pile
29	Bituminous paint is used to coat	2	external concrete surfaces	iron work under water	woodwork	compound walls
30	Borrow pits should preferably be located in	4	field on the left side of canal	field on the right side of canal	field on both sides of canal	central half width of the section of the canal
31	Bricks are burnt to	1	attain hardness and strength	colour and shape	get the correct shape	none of these
32	CBR means	3	Calibrated Bearing Ratio	Clear Bearing Ratio	California Bearing Ratio	Compact Bearing Ratio
33	Cement required for one cum M20 concrete is	2	2.0 bags	7.6 bags	2 bags	9 bags
34	Cement required for UCR masonry in CM 1:3 is	3	2.6 bags / cum	4.1 bags / cum	3.0 bags / cum	4.9 bags / cum
35	Classification of bridges according to loading are	4	class 'AA' bridges	class 'A' bridges	class 'B' bridges	all of these
36	Coefficient of friction is less when the pavement surface is	4	rough	dry	smooth and dry	smooth and wet
37	Contour interval is	1	the horizontal distance between two consecutive contours	the vertical distance between two points on same contour	the horizontal distance between two points on same contour	none of these

38	Contracts in Govt. are entered between contractor and	1	Executive Engineer	Superintending Engineer	Chief Engineer	none of these
39	Cross check for steel reinforcement of RCC slab is required to be done by	3	Executive Engineer of the concerned division	Deputy Engineer of concerned sub division	Deputy Engineer of any sub division not concerned	all of these
40	Cross cut saw is used for	2	cutting soft stones	cutting hard stones	cutting large blocks of stones	dressing stones
41	Curing of concrete is done to have	2	better appearance	become strong & durable	optimize use of cement content	none of these
42	Dado is usually provided in	2	dinning halls	bath rooms	living rooms	verandah
43	Distemper is used on	3	brick walls	plastered surfaces exposed to weather	plastered surfaces not exposed to weather	all of these
44	Due to temperature effect ----- are developed in concrete work	1	cracks	energy easements	both of these	none of these
45	Dumpy level is most suitable when	3	the instrument is to be shifted frequently	fly levelling is being done over long distance	many readings ae to be taken from a single setting of the instruments	all of these
46	Early attainment of strength in rapid hardening cement is mainly due to	3	gypsum	fine grinding	tricalcium silicate	tricalcium aluminate
47	Enoscope is used to find	2	average speed	spot speed	space-mean speed	time-mean speed

48	Factors affecting alignment of roads are	4	need of traffic	purpose and class of road	curve	all of these
49	Failure of the stability of slopes, generally occurs along	3	slip plane	horizontal surface	a curved surface	all the surfaces
50	Fluid is a substance which offers no resistance to change of	3	pressure	flow	shape	volume
51	For a number of columns constructed in a row, the type of foundation provided is	4	footing	raft	strap	strip
52	For colgrout masonry the water cement ratio is about	3	0.35	0.50	0.84	0.2
53	For determination of average annual precipitation in a catchment basin, the best method is	3	arithmetical method	Thiessen's mean method	Isohyetal method	none of these
54	FRPs means	2	fixed-reinforced plastics	fiber-reinforced plastics	final-reinforced plastics	fiber-ready plastics
55	Give full form of CWC	4	Centurian Water Commission	Central Water Committee	Central Wake up Commission	Central Water Commission
56	Gneiss is chemically classified as	3	calcareous rock	argillaceous rock	silicious rock	none of these
57	Grade of high strength defamed bar generally used is	2	Fe 1000	Fe 415	Fe 1400	Fe1200
58	Grader is used mainly for	4	trimming and finishing	shaping and trimming	finishing and shaping	finishing, shaping and trimming
59	Grage of high strength defomed bar generally used is	2	Fe 1000	Fe 415	Fe 1400	Fe1200
60	HFL full form is	1	Highest Flood Level	His Flood Level	Highest Flooding Length	Highest Flood Latitude
61	HPC is short form of	3	Heavy Pipe Culvert	High Pipe Culvert	Hume Pipe Culvert	Hume Pipe Cooling
62	Hydrograph is graphical representation of	1	Runoff and time	seepage and time	rainfall and time	discharge and time

63	Hydrology helps in	4	predicting maximum flows	deciding the minimum reservoir capacity	forecasting the availability of quantity of water at reservoir site	all of these
64	Hydrostatic pressure on a dam depend upons it	4	length	depth	shape	both depth & shape
65	IMD is short form of	2	Indian Medical Department	Indian Meteorological Department	Indian Material Draft	Indian Metro Department
66	In case of multistoreyed buildings, the forms to be removed first are	1	sides of beams and girders	column forms	bottom of beams and girders	all of these at the same time
67	In Deccan trap ------ basalt is seen	2	sedimentary	amygdaloidal	metamorphic	none of these
68	In steady flow of a fiuid, the total acceleration of any fluid particle.	3	can be zero	is never zero	is always zero	is independent of co-ordinates
69	Intersection method of detailed plotting is most suitable for	3	forests	urban areas	hilly areas	plains
70	Irrigation canals are generally aligned along	4	contour line	valley line	straight line	ridge line
71	Jumper is a tool used for	2	testing of stones	quarrying of stones	dressing of stones	none of these
72	Los Angeles testing machine is used to conduct	1	abrasion test	impact test	attrition test	crushing strength test
73	Low percentage of tricalcium silicate and high percentage of dicalcium silicate in cement result in	4	rapid hardering	high early strength	high heat generation	none of these
74	Mannings formula is used for calculating	2	volume of earthwork in canal	velocity of flow in open channel	velocity gradient in open channel	none of these

75	Manning's formula is used for calculating	2	volume of earthwork in canal	velocity of flow in open channel	velocity gradient in open channel	none of these
76	MCM is short form of	2	Meter Cubic Meter	Million Cubic Meter	Metric Cubic Meter	Million Cubic Mass
77	Member which is placed horizontally to support common rafter of a sloping roof is	2	cleat	purlin	batten	strut
78	Minimum compressive strength of OPC cement 43 grade at 28 days is	1	430 kg / cm^2	330 kg / cm^2	230 kg / cm^2	43 kg / cm^2
79	Minimum shoulder width as per the recommendation of IRC for major district roads, two lane is	1	1 m	1.5 m	2 m	2.5 m
80	Minimum thickness of wall where single Flemish bond can be used is	1	one brick thick	one and a half bricks thick	two bricks thick	half brick thick
81	Most commonly used pump for lifting water in water supply mains is	1	axial flow pump	reciprocating pump	rotary type pump	centrifugal pumps
82	Most suitable material for highway embankments is	3	organic soil	sand	silts	clays
83	Movable barrier of shutter component is	4	style	panel	rail	all of these
84	OMC is short form of	4	Other Moisture Content	Obey Moisture Content	Optimum Moisture Clarity	Optimum Moisture Content
85	Ornamental moulded course placed on the top of a wall, is	1	Cornice	Coping	Frieze	Lintel
86	Penetration test on bitumen is used for determining its	1	grade	viscosity	ductility	temperature susceptibility
87	Pitot tube is used for measurement of	3	Pressure	Flow	Velocity	Discharge
88	PPM means	1	Part Per Million	Part Per Meter	Per Part of Million	Part Per Mass
89	Pug mill is used for	2	preparation of clay	moulding of clay	drying of bricks	burning of bricks
90	Queen closer may be placed	3	in header course	in stretcher course	in header course next to first brick	in stretcher course next to first brick

91	Raft foundations are used for	4	Providing increased area of foundation over poor bearing capacity of soil	Spanning over small soft or loose pockets	Counter acting the hydrostatic effect	All of these
92	Refrogression of the bed level of a river down stream a weir, occurs due to	3	heavy impact of water	increase of the bed level	less percentage of silt	soft soil strata
93	Reynold number is the ratio of initial force and	1	viscosity	elasticity	gravity	surface tension
94	Ribbed for steel is not hooked because	3	its strength is reduced while bending	bond strength is not required for RCC	bond strength is developed due to ribs	none of these
95	River gauging data is useful for ----	1	Flood Management	Population Calculation	Forestation	Rural Development
96	Roads are classified based upon the different areas of the country as	4	national highway	state highway	district road	all of these
97	Safe bearing capacity of black cotton soil is	3	10 to 25 T/m^2	15 to 30 T/m2	5 to 15 T/m2	30 to 40 T/m2
98	Scrapers are classified as	4	single engine	twin engine	multy engine	all of these
99	Self cleansing velocity is	3	velocity at dry weather flow	velocity of water at flushing	velocity of which no accumulation remains in the drains	velocity of water in a pressure filter
100	Sieve analysis of sand is done to find out	2	bulking of sand	fineness modulus of sand	particle size of sand	none of these

101	Slump test of concrete is a measure of its	2	impact value	consistency	tensile strength	compressive strength
102	Soils are -------- phase materials	2	One	Three	Two	Ten
103	Some common sedimentary rocks are	4	chalk	coal	clay stone	all of these
104	Steel in RCC beam takes	3	dead load	live load	tension	compression
105	Steel strands used for post tensioning.	1	tendon	deck	pier	all of these
106	Stopping sight distance is always	1	less than overtaking sight distance	equal to overtaking sight distance	more than overtaking sight distance	none of these
107	Stream lines and path lines always coincide in case of	2	steady flow	laminar flow	uniform flow	turbulent flow
108	Structure constructed over the road for canal water flow is called as	3	Bridge	Siphon	Aqueduct	None of these
109	System of submission of B-1 tender is	2	one envelope	two envelope	three envelope	
110	The aggregates required for one km. length of W.B.M. road per meter width and for 10 mm thickness is	2	8 m^3	10 m^3	12 m^3	15 m^3
111	The arrangement of supporting an existing structure by providing supports underneath, is known as	2	Shoring	Underpinning	Jacking	Piling
112	The asbestos cement pipes are generally laid	2	horizontally	vertically	at an angle of 30	at an angle of 60
113	The atmospheric pressure at sea level is	4	103 kN/m2	10.3 m of water	760 mm of mercury	All of these
114	The bearing capacity of black cotton soil is measured as	3	in kg	in ton	in T/m^2	all of these
115	The brick laid with its breath parallel to the face of a wall is known as	1	header	stretcher	closer	none of these
116	The coefficient of friction between the concrete and soil is	3	0.20 to 0.25	0.25 to 0.30	0.30 to 0.35	0.35 to 0.50
117	The concrete slump recommended for beams and slabs is	3	25 to 50 mm	25 to 75 mm	30 to 125 mm	50 to 100 mm
118	The crushing strength of burnt brick II class shall be minimum of	2	200 kg / cm^2	35 kg / cm^2	400 kg / cm^2	350 kg/cm^2
119	The current meter is used to measure the	1	Velocity	depth	discharge	pressure
120	The design life of bridge is considered to be	2	3 years	100 years	150 years	200 years
121	The difference in level between the top of a bank and supply level in a canal is called	2	berm	free board	height of bank	none of these

122	The distemper is used on	3	brick walls	plaster surfaces exposed to whether	plaster surfaces not exposed to whether	all of these
123	The end supports of a bridge superstructure are known as	3	pier	wing walls	abutments	approaches
124	The examples of earth moving machines are	4	scrapers	grader	dozer	all of these
125	The examples of earth moving machines are	1	scrapers	Concrete Mixer	Road Roller	all of these
126	The exterior angle between outer faces of a wall, is known as	3	Turn	Junction	Quoin	All of these
127	The foundations are placed below ground level, to increase	3	Strength	Workability	Stability of structure	All of these
128	The full form of ODD	1	Optimum Dry Density	Optical Dry Density	Optional Dry Density	Optimum dew density
129	The full form the SWF is	2	slant wave flume	standing wave flume	stone wind flume	starting wave flume
130	The highest point on road surface is called	1	crown	camber	gradient	berm
131	The hydraulic grade line is	2	never above the energy grade	always sloping downward in the direction of flow	never below the energy grade	all of these
132	The increase in the upstream water elevation resulting from an obstruction to flow	3	tail water	sub water	Backwater	riser
133	The instrument used for measuring the area on map is known as	3	speedometer	ammeter	planimeter	none of these
134	The intermediate support of a bridge superstructure are known as	2	abutments	piers	wing walls	approaches
135	The least count of a 4 m leveling staff is	1	5 mm	10 mm	15 mm	1 mm
136	The length of road visible ahead to the driver at any distance called as	3	front distance	back distance	sight distance	none of these
137	The liquid and plastic limits predominantly exist in	3	gravel soils	silty soils	clay soils	sandy soils
138	The maximum design gradient for vertical profile of a road is	1	ruling gradient	limiting gradient	minimum gradient	none of these
139	The member which is placed horizontally to support common rafter of a sloping roof is	1	purlin	cleat	batten	strut

140	The minimum design speed for hairpin bends in hill roads is taken as	1	20 kmph	30 kmph	40 kmph	50 kmph
141	The minimum recommended diameter of sewer is	3	5 cm	10 cm	15 cm	20 cm
142	The most powerful explosive used in blasting is	2	blasting power	dynamite	gun cotton	cordite
143	The piece of a brick cut with its one corner equivalent to the half the length and half the width of a full brick is known as	3	queen closer	bevelled closer	king closer	half king closer
144	The pore pressure developed due to soil mass due to moisture is measured with	2	earth pressure cells	piezometers	uplift pressure cells	none of these
145	The process of making the back ground rough before plastering is	2	dubbing	hacking	blistering	peeling
146	The purpose of COT in the earthen dam is to	3	Irrprove the stability of dam	increase density of hearting	decrease the seepage from reservoir	all of these
147	The removal of excess air after placing concrete helps in increasing the strength of concrete by	1	15 to 20%	20 to 30%	30 to 50%	50 to 70%
148	The ruling design speed on a National Highway in plain terrain as per IRC recommendations is	3	60 kmph	80 kmph	100 kmph	120 kmph
149	The shinkage in a concrte slab,	2	causes sheat cracks	causes tension cracks	causes compression cracks	does not cause any cracking
150	The size of sample cube for concrete is	4	10 cm x 10 cm x 10 cm	12 cm x 12 cm x 12 cm	20 cm x 20 cm x 20 cm	15 cm x 15 cm x 15 cm
151	The slenderness ratio for masonry walls should not be more than	3	10	20	30	40
152	The slump test of concrete is used to measure its	4	impact value	homogeneity	tensile strength	workability
153	The surface runoff is the quantity of water	4	absorbed by soil	intercepted by buildings and vegetative cover	required to fill surface depressions	that reaches the stream channels
154	The thickness of a sharped crested weir is kept less than	2	one third of the height of water on the sill	one half of the height of water on the sill	one forth of the height of water on the sill	two third of the height of water on the sill
155	The type of footing, which is used to transmit heavy loads through steel column is	2	raft foundation	grillage foundation	well foundation	isolated footing
156	The type of pile which is driven at an inclination to resist inclined force is known as	3	friction pile	sheet pile	batter pile	anchor pile

157	The type of valve which allows water to flow in one direction but prevents its flow in the reverse direction is	1	reflux valve	sluice valve	air relief valve	pressure relief valve
158	The unit weight of per cubic meter of mild steel is	3	7725 kg / m^3	7250 kg / m^3	7850 kg / m^3	8000 kg / m^3
159	The unit weight per cubic meter of water	1	1000 kg / m^3	800 kg / m^3	1500 kg / m^3	900 kg / m^3
160	The vertical cutting of river bed is called	2	afflux	scour	water way	none of these
161	The vertical distance between the wall plate and top of the ridge is called ……of roof	1	rise	pitch	template	gable
162	Water cement ratio of water to cemnt	1	by weight	by volume	none of these	
163	Water supply system includes	4	digging a well for water	construction of dams	construction of canals	entire arrangement from source to distribution
164	When a canal and a drainage approach each other at the same level, the structure so provided is	3	an equeduct	a syphon	a level crossing	inlet and outlet
165	When a canal is carried over a natural drainage, the structure provided, is known as	2	syphon	aqueduct	super passage	syphon aqueduct
166	Which of the following is a non-recording rainguage	3	tipping bucket type rainguage	Simons rainguage	Steven's weighting type rainguage	floating type rainguage
167	Without issue of notificaion under section 4 (1) of L.A. Act., lands can be taken in possession by private negotiation is	2	thue	FALSE	-	--